# Knowledge Management in Genomics: The Role of Data Provenance

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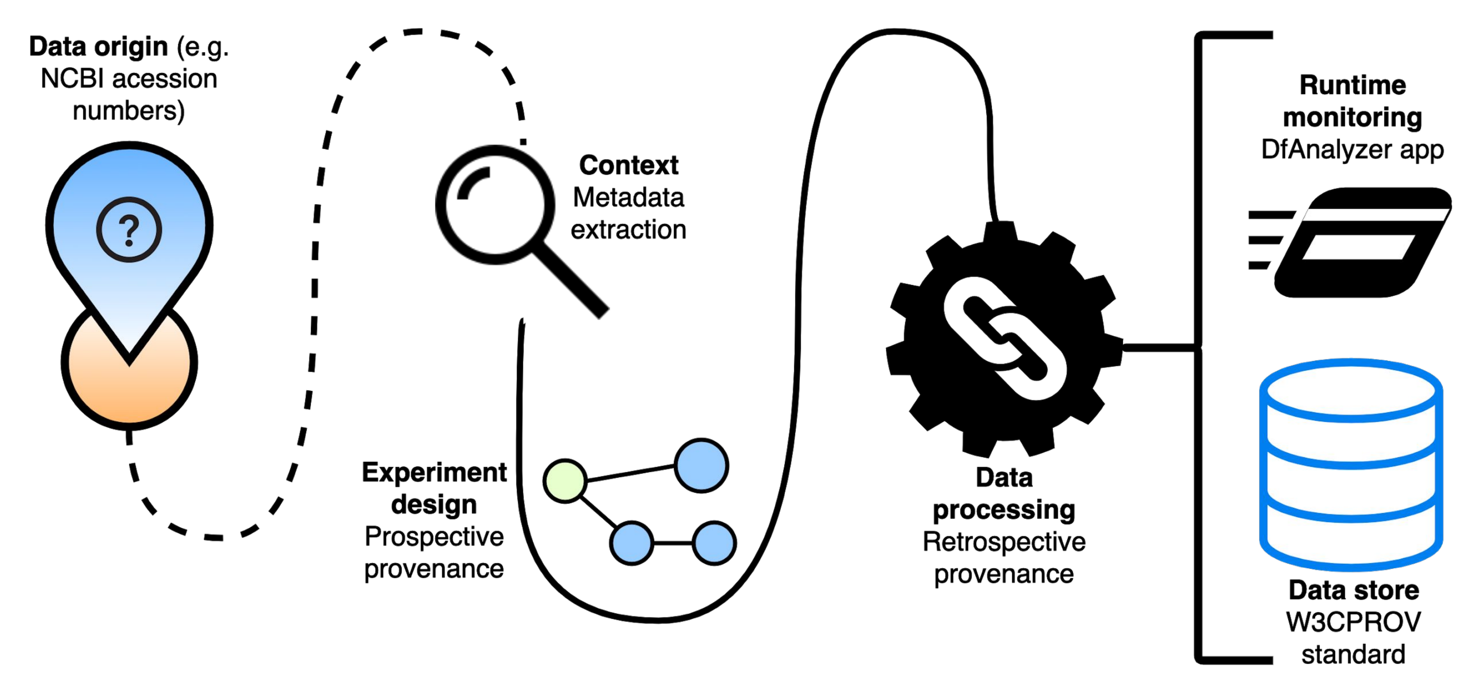
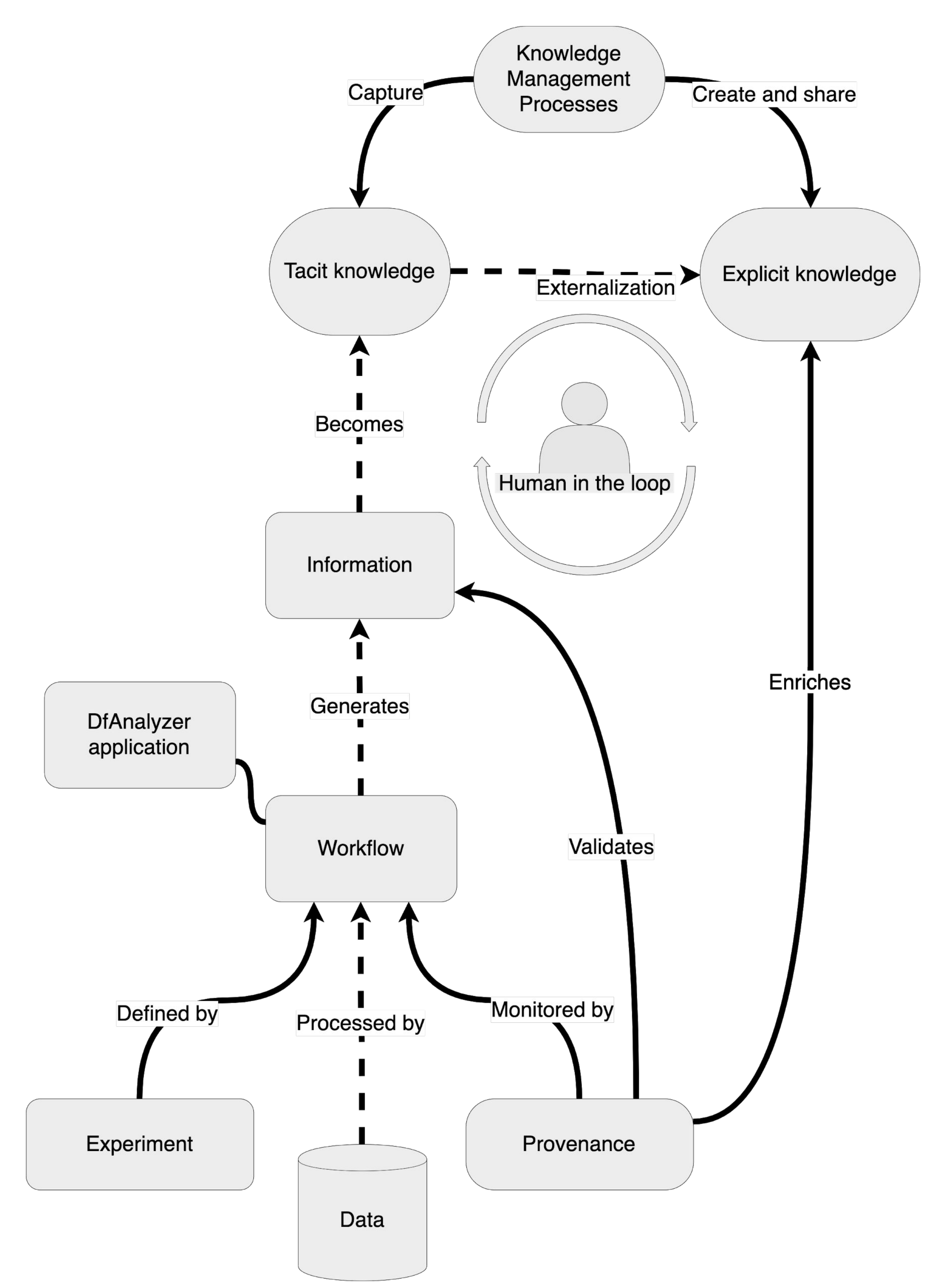
## Genomics as a data-driven field

- With the rise of high throughput sequencing technologies, genomics data is growing at exponential rates.[1]
- Research in genomics is highly dependent on public data repositories.[2]
- *In silico* experiments make use of extensive computational transformations of data.[3]
- Proper management of data is required for a study's reproducibility.[4, 5]
- Describing the **lineage of data** facilitates capturing, creating and sharing knowledge around scientific experiments.[6]

## Provenance: if these data could talk[7]



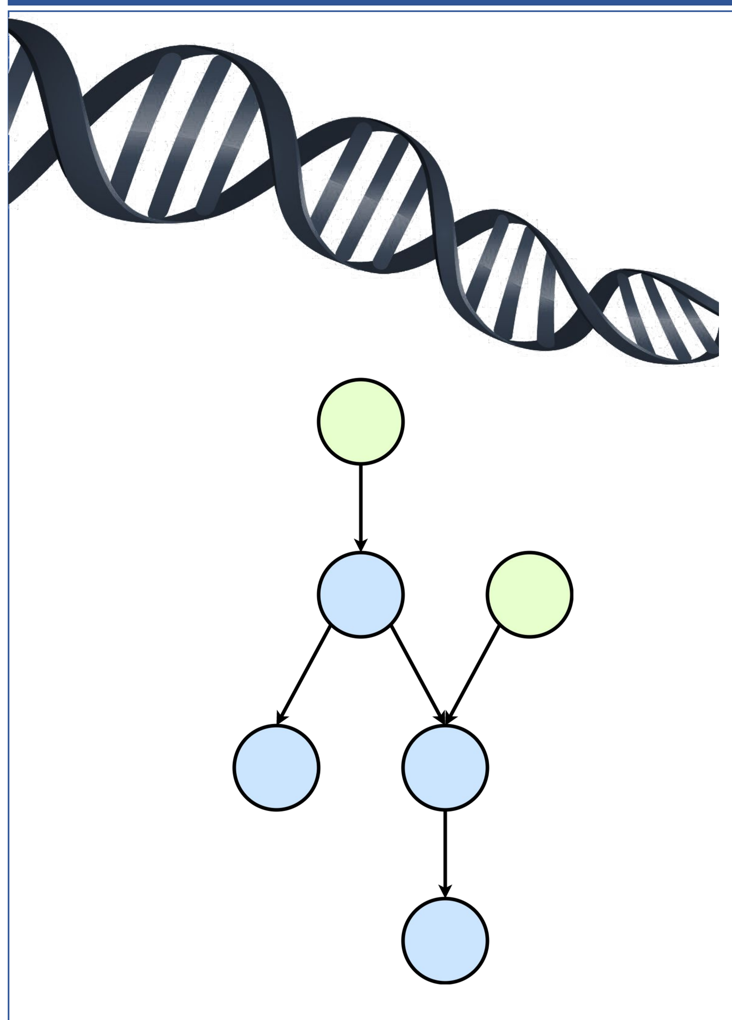

## Knowledge Management flow




## Dataflow concepts in DfAnalyzer[8]

| Concept | Definition |
|---|---|
| data element | $e = (v_1, v_2, \dots, v_\alpha) \therefore$ $v_i$ is an attribute value for each $a_i$ in $A$ $\wedge\, a_i = (name, type)$ |
| data collection | $c = \{e_1, e_2, \dots, e_\beta\}$ |
| dataset | $s \equiv C$ |
| data transformation | $t = (S_{input}, S_{output})$ |
| data dependency | $\varphi = (s, t_{previous}, t_{next})$ |
| dataflow | $D_F = (T, S, \Phi)$ |

## Knowledge management processes in a provenance-aware genomics experiment



Prospective provenance: dataflow modeled as DAG

### Annotation and comparison workflow

167 assembled *Synechococcus* genomes from GenBank are loaded as Python objects with the Abacat library.

Data transformations include:

- Assembly statistics
- Genome completeness and contamination
- Gene calling
- Annotation with different databases
- Pairwise comparisons between genomes

1. **Model workflow**
2. **Run with DfAnalyzer**
3. **Query database**

```
SELECT task_id, accession, ref_db, evalue, hits
FROM annotation WHERE refdb = 'COG' LIMIT 5;
```

| Task ID | Accession | Ref db | E-value | Hits |
|---|---|---|---|---|
| 54 | GCA_000332275.1 | COG | 10e-6 | 2304 |
| 55 | GCA_000155595.1 | COG | 10e-6 | 1830 |
| 56 | GCA_000019485.1 | COG | 10e-6 | 2142 |
| 57 | GCA_001693275.1 | COG | 10e-3 | 3144 |
| 58 | GCA_001693295.1 | COG | 10e-3 | 2934 |


Retrospective provenance: provenance data stored in relational database

Each row a data element $e$; Each table a data collection $c$

**Provenance databases become a resource for knowledge discovery**

We design a genomics experiment with arbitrary parameters. → Workflow modeling describes data transformations following W3CPROV format → Provenance monitoring during runtime provides a mean to effectively query different parameter responses → Provenance-aware applications as a framework for KM in genomics

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